SRI HARI N U

Software Engineer II – Backend, Distributed Systems & Applied AI

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PROFESSIONAL SUMMARY

Software Engineer with **4+ years of experience** at J.P. Morgan, building high-throughput, scalable distributed systems for **Equity Derivatives Trade Management**, with domain knowledge of regulatory assessment, trade lifecycle, corporate actions, and derivatives across Equity, FX, Credit and Commodities.

Specialized in Event Driven Architecture, Microservices, Distributed Systems, REST APIs, Performance Optimization, Data Pipelines, System Design (LLD/HLD) and Agentic AI workflows.

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Python, FastAPI, SQL, React

Data & Streaming: Kafka, Spark (PySpark), Avro, Parquet, Pandas

Database & Caching: PostgreSQL, Sybase, MySQL, pgvector, GemFire (Apache Geode)

Cloud & Infrastructure: AWS - EC2, ECS, S3, SQS, SNS, Lambda, Glue, Athena, Step Functions, Terraform

DevOps: Docker, Kubernetes, Jenkins, Spinnaker, Git, Datadog

Testing: Pytest, Playwright, JUnit, Mockito

AI Tools: Claude Code, LangChain, RAG Pipeline, MCP

EXPERIENCE

J.P. Morgan Chase & Co. | *Bengaluru, India*

Software Engineer II | *Feb 2024 – Present*

- Built and maintained equity derivatives trade management core services processing **400K+ events** daily across a hybrid-cloud architecture distributed across North America, EMEA, APAC regions, feeding downstream systems through AMPS messaging, IBM MQ, and centralized file transfers.
- Implemented **Common Domain Model** (CDM)-based Digital Regulatory Reporting (DRR) for equity derivatives, contributing to the **firm's FINOS Adoption Achiever Award**.
- Delivered regulatory rewrites across **six regimes**—SEC, FCA, CFTC, ASIC, HKMA and JFSA - by implementing regulatory rules, integrating reference data, and enriching existing trades and positions representing **over a trillion-dollar in market value**, achieving approximately **97% report acceptance** by DTCC.
- Built **a trade-data pipeline** into the **firm-wide data lake** using Glue, PySpark, Athena, S3, ECS, and SQS. Implemented retries and end-of-day reconciliation against source audit records enabling **the data consistency and idempotency** in the data lake.
- Reduced CDM transaction-model creation time **by 5x (10 min → 2 min)** for composite trades containing 5K+ positions by optimising data transformers, mapping logic, and model construction, **improving throughput and preventing processing backlogs** during peak traffic.
- Redesigned end-of-day regulatory report processing using data segregation, batching, parallel processing, concurrent task execution, and API optimization, increasing throughput and generation time **by 86% (3 hrs → 25 min)**, **eliminating compliance SLA breaches**.
- Developed a **RAG assistant** using React, FastAPI, an internal ADK (Fusion SDK) built on LangChain, secured with OAuth 2.0 and JWT access-token validation. Grounded responses in Jira, Confluence, video transcripts and internal documentation to help Operations and Engineering teams **navigate over 45+ workflows**.
- Built **multiple copilot agents** to orchestrate **15+ components** across multiple technology stacks in local development environments, improving development speed by approximately **20%**.
- Built **an automated end-to-end regression suite** (Pytest, AWS SDKs, Playwright) for testing in pre- and post-deployment stages in CI/CD, reducing manual regression effort and **saving 8+ developer-hours weekly.**

Consultant | *Sep 2022 – Jan 2024*

- Built a high-throughput US equities **trade-feed consumer** with a **custom-optimized Avro deserializer**, transformers, and legacy reconciliation mappings, achieving a 99% clearing rate.
- Automated **end-of-day reconciliation** using scripts and Control-M(scheduler); ingested files into Sybase tables and enriched records through multi-table SQL joins, **eliminating manual position-break checks**.
- Built a **custom Kafka Connect source connector** for insert-only CDC from Sybase event tables, using timestamp-based SQL polling to publish new events to Kafka for downstream consumers.
- Streamlined **CI/CD pipelines** (Terraform, Docker, Jenkins, Spinnaker, Sonar, Snyk) with gated code security and vulnerability scanning, cutting deployment time **by 60%**.

EDUCATION

B.E. in Computer Science & Engineering, Visvesvaraya Technological University, Belgaum | *2018 – 2022*

CERTIFICATIONS & AWARDS

AWS Certified Cloud Practitioner (2024) | FINOS Adoption Achiever Award (2024)